IT342 Phase 1 – Implementation Summary

Project Name: Halaman

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Course & Section: IT342-G2

GitHub Repository: https://github.com/KennyDevz/IT342-Devibar-Halaman.git

Final Commit: f0ea8ef0b9a21e2f191277d3227dc0e0c62962c1

User Registration

Registration fields used:

- First Name, Last Name, Email, Password, Confirm Password

Validation process:

- All fields are required and validated on both frontend (React) and backend (Spring Boot @Valid annotation)
- Email must follow a valid format
- Password must be at least 8 characters
- Confirm password must match password

How duplicate accounts are prevented:

- Before saving, the system calls userRepository.existsByEmail() to check if the email is already registered
- If duplicate is found, returns error code DB-002 with message "Email already registered"

How passwords are stored securely:

- Passwords are hashed using **BCrypt** with salt rounds of 12 before being stored in the database
- The raw password is never stored only the hash in the password_hash column

User Login

Login credentials used:

- Email and Password

How the system verifies users:

- Credentials are passed to Spring Security's AuthenticationManager
- It loads the user by email via UserDetailsService and compares the submitted password against the stored BCrypt hash
- If invalid, returns error code AUTH-001 with message "Invalid credentials"

What happens after successful login:

- Server generates a JWT access token (valid 24 hours) and a refresh token (valid 7 days)
- Tokens are returned in the response and stored in the browser's localStorage
- User is redirected to the dashboard

Database Table

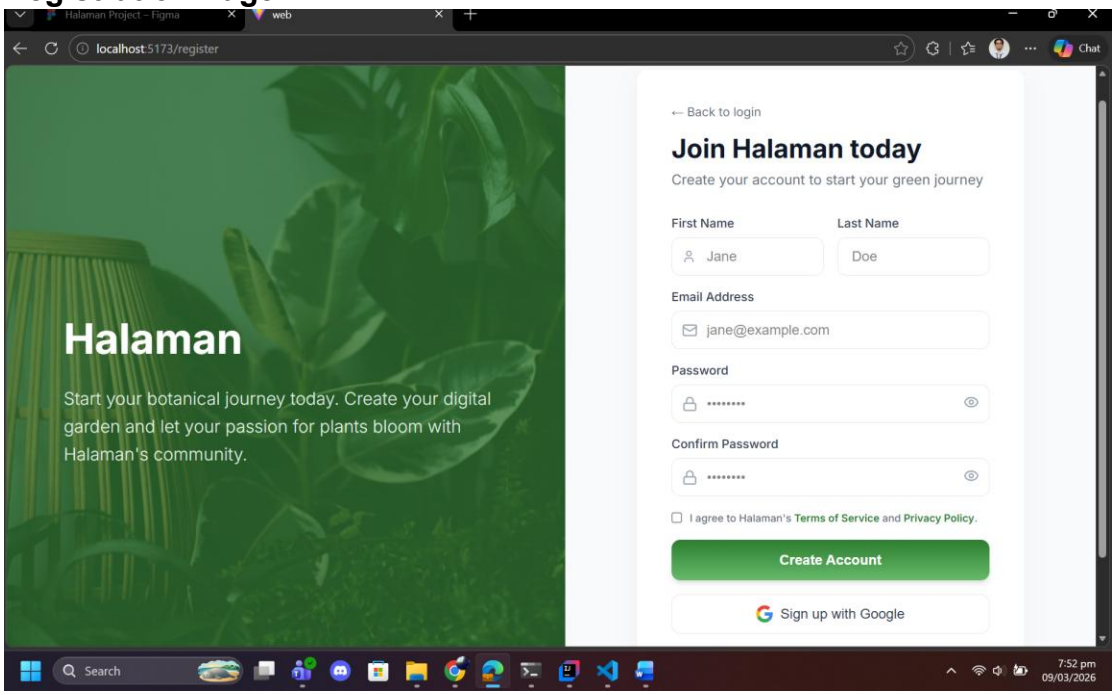
Column	Type	Description
user_id	UUID	Primary key, auto-generated
email	VARCHAR(255)	Unique, used for login
password_hash	VARCHAR(255)	BCrypt hashed password
google_id	VARCHAR(255)	For Google OAuth (future)
first_name	VARCHAR(100)	User's first name
last_name	VARCHAR(100)	User's last name
role	VARCHAR(50)	USER or ADMIN
created_at	TIMESTAMP	Account creation time

API Endpoints

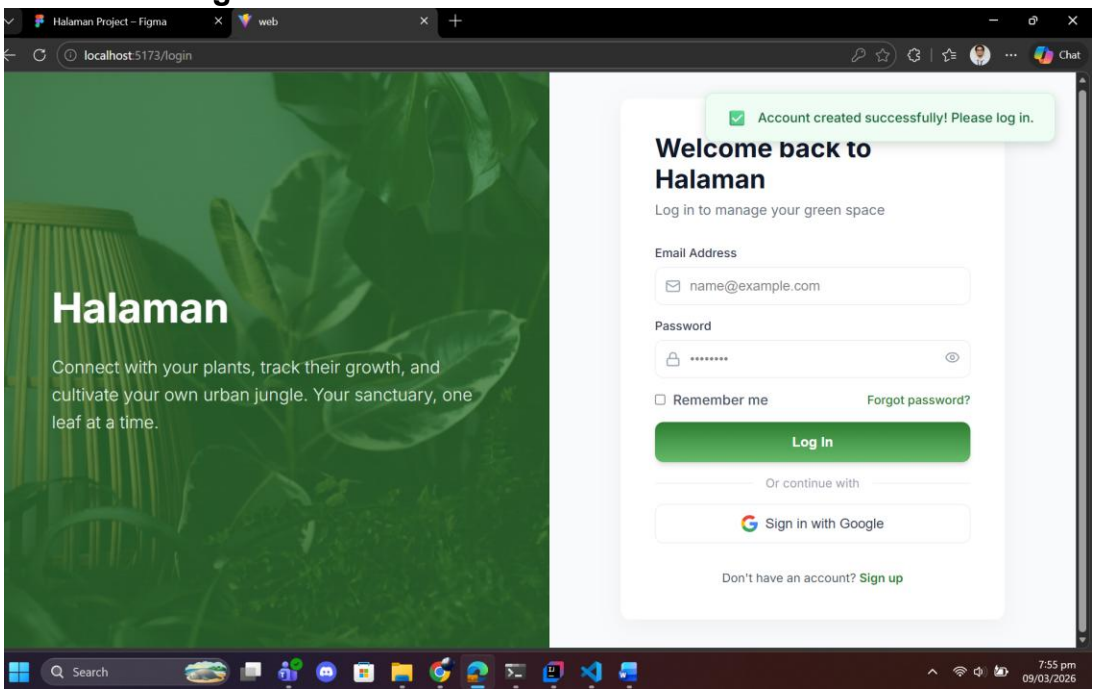
Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Login user
GET	/api/auth/me	Get current logged-in user details

SCREENSHOTS

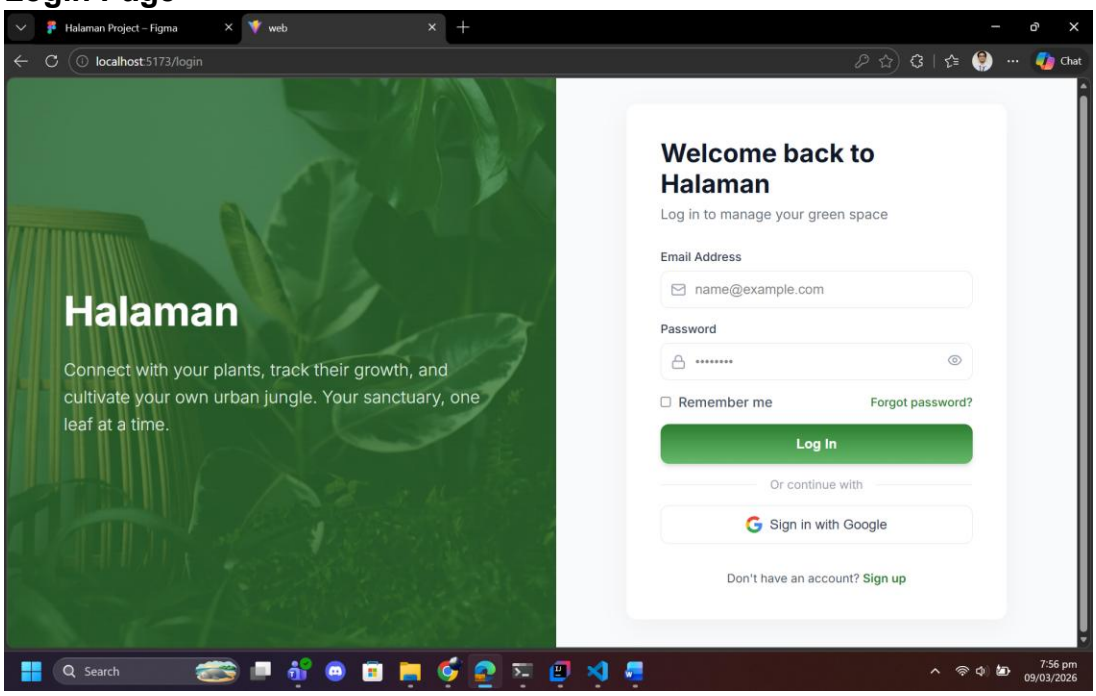
1. **Registration Page**



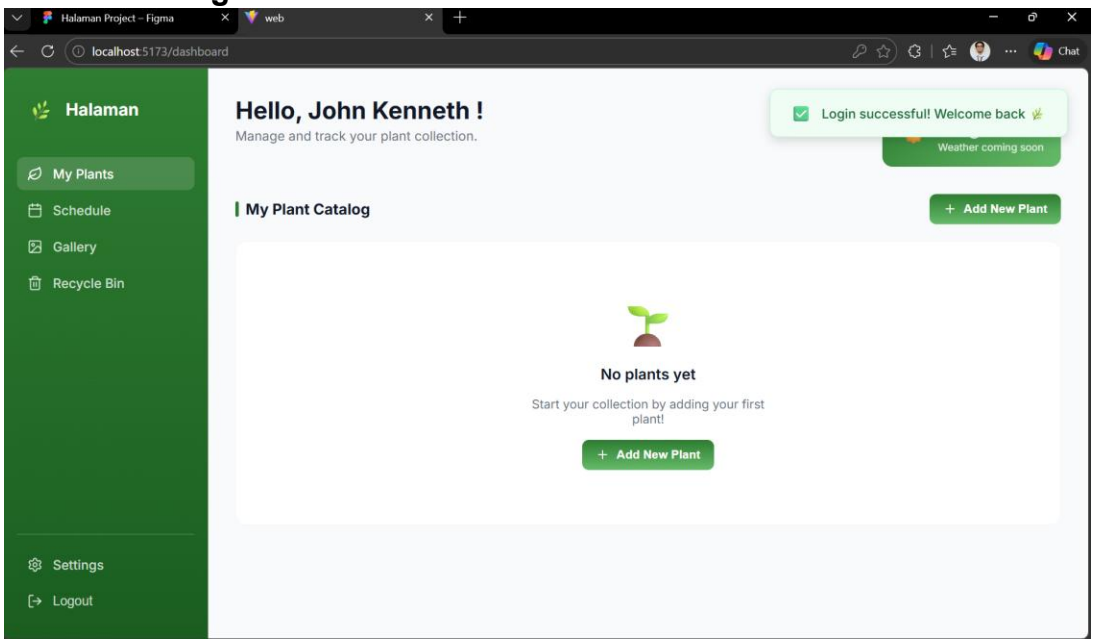
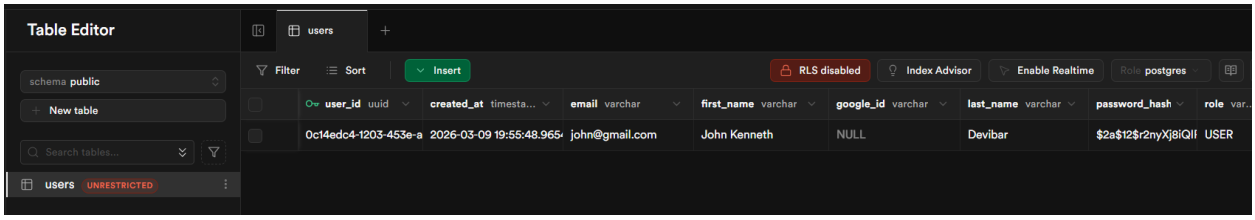
2. **Successful registration**



3. Login Page



4. Successful login



5. Database record